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Geospatial Prediction of Geochemical Properties of Produced Water Samples: Application of Dual Kriging-Gradient Boosting Method

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Abstract

The geochemical assessment of produced water has a practical impact for operational decisions in oil and gas reservoir management. As a general practice, kriging and other geo-statistical interpolation methods are deployed to estimate values at unsampled location away from well control. A key disadvantage of spatial Gaussian process models (e.g., kriging) is their assumption of a linear relationship between response variable (water salinity) and the predictor variables (such as elemental concentration of Na, Cl, Ca, HCO_3, SO_4). Resulting predictions for spatial data analysis are discontinuous over space, and machine-learning (ML) algorithms tend to ignore geo-spatially coherent structure of input data. In order to overcome the limitation of Gaussian process models, kriging and gradient tree boosting algorithms were combined as a novel practical solution to ensure an optimum predictability of spatial and temporal geochemical variations. As a proof of concept, a historical geochemical dataset of produced water from national exploration and production areas was recovered from the USGS National Geochemical Database with an extended collection of measured values of chemical-physical parameters (TDS, density, pH), and elemental concentrations (Na, Ca, K, Mg, Ba, Sr, Cl, HCO3 and SO4). The USGS dataset was utilized to choose an appropriate "target" variable (such as TDS) and use remaining predictor variables as "features" which constitutes the input for supervised ML framework used. A detailed exploratory statistical data analysis for input features was performed using correlation matrix, analysis of probability distribution and quantile-quantile plots (Q-Q plot). Geochemical data analytics with Piper and Gaillardet end-member diagrams provided an automated characterization of recovered fluids for given stratigraphic formations. The application of coupled gradient-boosting and universal kriging methods improved the prediction for target variables that deviated from the assumed normal distribution (Gaussian assumption). This method will overcome inherent limitations of kriging method associated with non-Gaussian predictive variables, especially at unsampled locations. The proposed numerical technique represents a generalized geo-statistical tool to predict different geochemical parameters in the reservoir ahead of the drill bit.

Introduction

The oil and gas industry is generating significant volumes of produced water during production and operations phases throughout the lifetime of the field. The chemical composition of this *produced water* varies depending on factors such as the geological age, depth, and inherent geochemistry of the formation, as well as chemicals, additives and injected fluids used during production (Abdou et al., 2011). Geochemical fingerprinting of this oilfield water samples has significant value for practical applications within oil and gas reservoir management, such as tracking hydrodynamics of groundwater systems (Souid et al., 2023), reservoir compartmentalization (Birkle & AlRamadan, 2023), to assess the functionality of fractures and efficiency of frac jobs during hydraulic fracturing (Birkle & MaKechnie, 2021) and to avoid scaling tendencies during waterflooding (Birkle et al., 2023). As a general practice, geochemical analysis is performed on water samples collected at well head or bottom holes, i.e., as point measurements. Specific operational objectives, such as the identification of dynamic hydraulic systems by gradual increasing trends of formation water salinity, require the prediction of water properties, e.g. salinity, at a regional field scale. Hence, geo-spatially consistent prediction of geochemical properties such as total dissolved solids (TDS), elemental concentrations (e.g., Na+, Ca++), and pH at unsampled locations is essential for subsurface reservoir management, water chemistry monitoring, and operational decision making. By providing a continuous spatial understanding of formation water chemistry on a regional scale, such predictions have the potential to accurately characterize reservoir heterogeneity, and to ascertain hydraulic connectivity between wells. For operational purposes, engineers can optimize production and injection strategies if reliable spatial geochemical models are available at regional scale. For instance, predicting salinity or pH across the field assists in early detection of scaling or corrosion risks (Webb & Kuhn, 2004). With the application of numerical prediction for regional salinity and geochemical trends, a considerably reduced reliance on costly and time-consuming water sampling from every well is expected, including a significant potential to lowering analytical expenses while improving coverage and decision accuracy. From an environmental and regulatory perspectives, spatially continuous prediction is critical in ensuring safe operations near aquifers or sensitive areas.

To produce geo-spatially consistent predictions of geochemical properties (e.g., TDS, Na+, Ca++ or pH) at unsampled locations, one needs numerical methods that account for spatial autocorrelation, trends, uncertainty quantification, and ensure underlying chemical relationships (Humphrey et al., 2023). One of the most widely used approaches is geostatistical interpolation, particularly kriging (Journel & Huijbregts, 2004). Ordinary kriging provides unbiased estimates of a parameter by weighting nearby samples by fitting a variogram model. Kriging methods, while widely used for geochemical spatial interpolation, have several key limitations (Gentile, 2012). First, it assumes stationarity conditions, meaning that the spatial correlation structure (variogram) is consistent across the study area, which may not hold true for heterogeneous aqueous medium with varying geological or environmental conditions. Additionally, the kriging methodology struggles with non-linear relationships and complex interactions between variables. More recently, machine learning methods, such as gradient tree boosting (XGBoost) have been applied for geochemical gold grade in mining industry (Ibrahim et al., 2022). Within the sector of geotechnical technology, Random Forests regression has been employed to predict soil chemical properties (Siqueira et al., 2013) (Ho, 2024). These models can capture complex and nonlinear relationships between geochemistry and physical variables. They do not account for spatial structure unless geographic coordinates or engineered features are included explicitly as inputs. This simplistic formulation can lead to overconfident predictions in areas with strong spatial trends.

Methodology

In this paper, an aggregated universal kriging (Cao et al., 2013) and gradient tree boosting method (Nielsen, 2016) has been applied to geochemical datasets. This method has the ability to enhance geospatial predictions of geochemical parameters by leveraging the strengths of both methods while mitigating their individual limitations. Gradient boosting excels at modeling complex and non-linear relationships between geochemical data and environmental covariates (e.g., stratigraphic formation and lithology), while kriging explicitly accounts for spatial auto-correlation, providing robust interpolation and uncertainty quantification. By combining these approaches, the proposed hybrid model can achieve higher accuracy and robust prediction at unsampled locations, better than either method by itself could achieve. Mathematically, this aggregated method can be equivalently expressed as follows:

$$y = F_{XG}(\mathrm{x,s}) + b_{kg}(\mathrm{x,s}) + \epsilon$$

where y is the response variable (also known as label), $F_{XG}(\mathbf{x}, \mathrm{s})$ is the non-linear function with spatial component containing the predictor variables (also known as input features), $b_{kg}(\mathbf{x}, \mathrm{s})$ is the Gaussian process such as universal kriging and ϵ is the error term. In this case, both universal kriging and XGBoost regressor models are applied on the observed data to predict missing values at unsampled locations independently. In order to incorporate the total variance described within the sampling space, an aggregation function is deployed to combine results from both predictors. In this way, the limitations of kriging (assumption of linearity) and tree boosting algorithms (lack of incorporation of discontinuity over space) are overcome by creating a "dual" approach. The first step, in case of dual kriging-gradient tree boosting algorithm implementation, is to set the target coordinates in the regular grid data of "unsampled" location with the required prediction performance. Subsequently, the workflow leverages the gradient tree based boosting method, such as eXtreme Gradient Boosting (XGBoost) (Chen & Guestrin, 2015) to train a regression model on the observed data and predicts the missing values for the unsampled data using the trained model for a target variable such as TDS. The goodness of the trained regressor model is measured through the Mean Squared Error (MSE) term (average squared difference between actual and predicted values) and R^2 values. These values reflect the performance of the model to explain the variability of the target variable, such as predicted TDS at unsampled mesh grid points). Once a satisfactory level of confidence is reached (reasonable convergence values from MSE term and R^2 value), the predicted values are further leveraged toward universal kriging implementation by combining external drift variables. Finally, individual variances are weighted and aggregated as per defined rules, and thus the prediction variance of the dual methodology is decreased.

As per the practical software implementation of the proposed methodology, open-source python programming-based libraries have been leveraged. For both ordinary and universal kriging, *PyKrige* (kriging toolkit for python) libraries have been extensively used (Fahimuddin et al., 2024). As per the machine learning implementation for gradient tree boosting algorithm, extensive use of *scikit-learn* library has been ensured (Pedregosa et al., 2011). In addition, eXtreme Gradient Boosting (XGBoost) has been utilized as the component for gradient tree boosting algorithm implementation.

Case study: Prediction of TDS using dual kriging-gradient boosting method

The proposed methodology has been implemented in form of a representative "synthetic use case" by leveraging data from the openly available geochemical database of produced water from the U.S. Geological Survey (USGS), which contains geochemical and other information for 100,000+ produced water and other deep formation water samples of the United States from 300+ wells (Blondes et al., 2019). This dataset is an extensive collection of measured values of chemical-physical parameters (TDS, density, PH) and elemental concentrations (Na+, Ca++, K+, Mg+, Ba++, Sr++, Cl-, HCO_3- and SO4--) of produced water from historical samples which are essential for the understanding of the evolution and origin of formation water and fluid interaction with host rocks. As shown in Figure 1, the focus has been on USGS produced water data to support geochemical fingerprinting prediction, particularly in the context of hydrocarbon-bearing formations and stratigraphic units of the Smackover, Tuscaloosa, and Wilcox Groups. The spatial map corroborates that sampled wells are distributed across Louisiana, Mississippi, and Alabama, and hence, geo-spatially these data points are relatively sparse. On the right side of Figure 1, two Gaillardet End-Member Diagrams (Gaillardet et al., 1999) compare geochemical ratios to ascertain dominant geochemical processes and source minerals. These diagrams demonstrate that USGS produced water samples are likely to be influenced by evaporite dissolution (e.g., halite, gypsum, anhydrite), particularly in Smackover, Tuscaloosa, and Wilcox formations. This also discounts the fact that the samples have little influence from carbonate or silicate mineral weathering. Hence, it is possible to assert the notion that produced waters in this region are dominated by *Na-Cl* and *Na-Ca–Cl* chemistry, originated by evaporite dissolution during geological time, and this process governs the spatial and compositional variability of key geochemical parameters like TDS. This validation is important for building reliable models and hybrid spatial interpolators such as the proposed dual kriging-gradient tree boosting method. This ensures that model features reflect true hydrogeochemical processes, not just statistical artifacts.

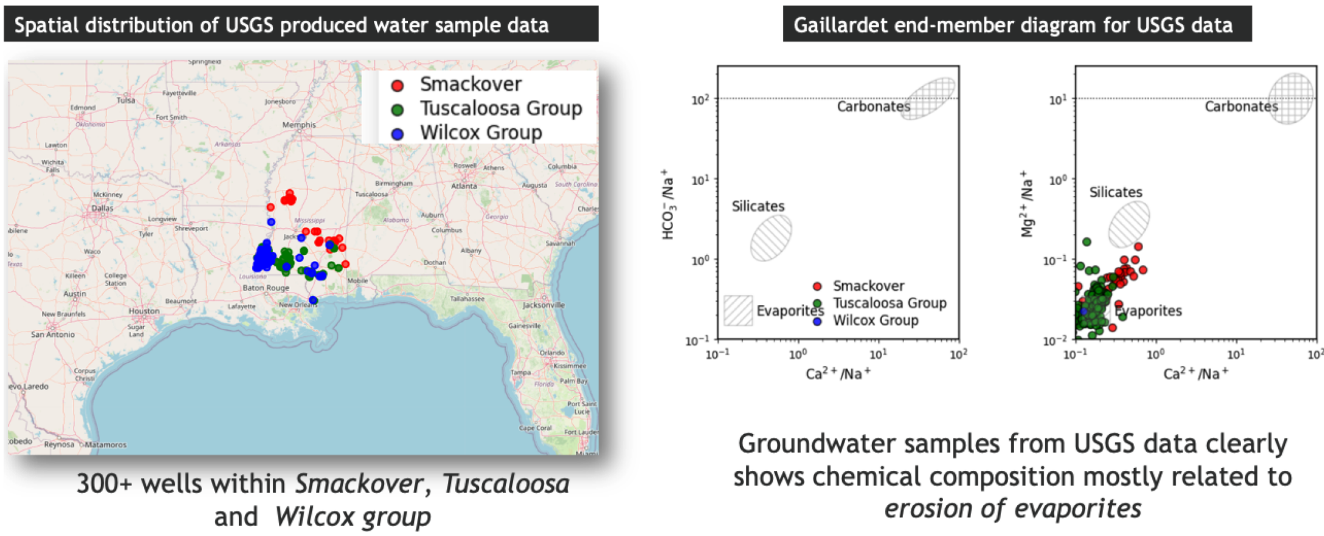

Figure 1—USGS produced water samples data: understanding formation water evolution, origin and interactions with host rocks

For a better understanding of USGS water samples, a detailed exploratory data analysis has been performed. The Piper diagram is a standard geochemical tool that is used to classify water types based on their major cation and anion compositions (Pepin et al., 2014). As seen in Figure 2, the studied water samples are mainly related to the *Na-Cl* and *Na-Ca–Cl* facies, indicating that produced waters are characterized by high sodium, calcium and chloride concentrations. This identified composition trend in the Piper diagram is strongly correlated with evaporite dissolution processes (already seen from Gaillardet diagrams), due to the water-rock interaction with halite (*NaCl)* and gypsum. Observations from both plots (Piper diagram and Gaillardet end-member diagrams) reconfirm calcium as a major contributor to the overall dissolved solids content in these produced waters. The observed *Ca–TDS* relationship confirms that evaporite mineral dissolution (e.g., gypsum, anhydrite) is a dominant process influencing water chemistry, especially in Smackover and Tuscaloosa reservoirs.

For the implementation of the proposed hybrid method, several key python programming steps with appropriate parameters were applied. For the gradient tree boosting part, *xgboost* library and function *XGBRegressor* were leveraged. To train the model, a split of 70/30 was used for training and test data set, and to ensure model accuracy, both Mean Squared Error (MSE) and R^2 values were applied using scikit-learn library. Universal kriging was implemented by utilizing PyKrige library function *UniversalKriging* with spherical variogram. Statistical measures, such as variance calculation was performed with native python functions. Finally, predicted results were plotted as isoline maps on an equally spaced regular grid using python library matplotlib function *contourf,* ensuring proper interpolation of sample points across the chosen 2D domain.

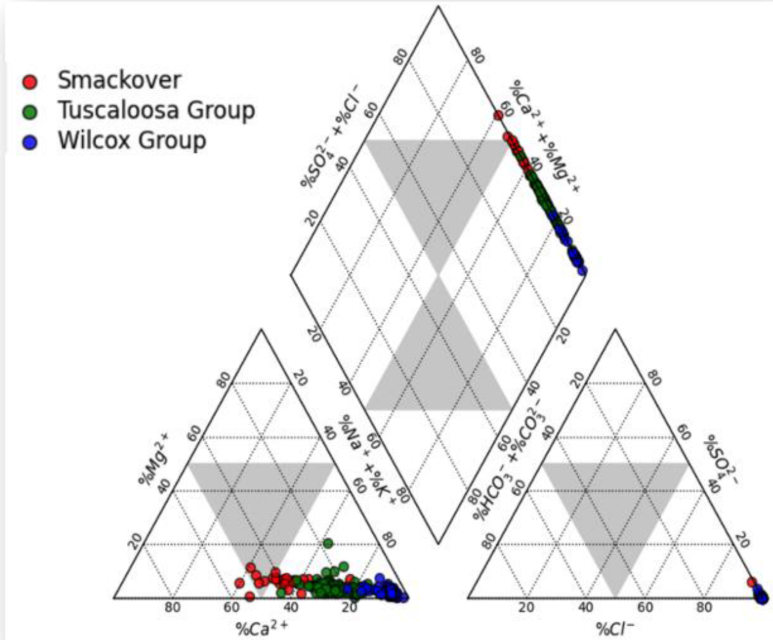

Figure 2—Piper diagram (classification of water types based) of the USGS produced water samples data

Results and Discussion

In this paper, a dual kriging-gradient tree boosting method has been proposed for predicting geochemical properties at unsampled locations. Historical geochemical data of produced water samples were taken from single well locations. As both methods, traditional kriging and machine learning based gradient tree boosting, have specific limitations in their applicability (see Introduction chapter), the rationale is to utilize a hybrid method to improve overall predictability of geo-spatially challenging non-Gaussian parameters (i.e., Ca++ concentration). The comparative analysis in Figure 3 underscores the remarkable improvement of the geochemical prediction for total-dissolved-solids (TDS) with the hybrid dual workflow (kriging-gradient tree boosting). The methodology was especially effective in areas without geochemical data, and for locations with distant sampling sites and known geochemical attributes.

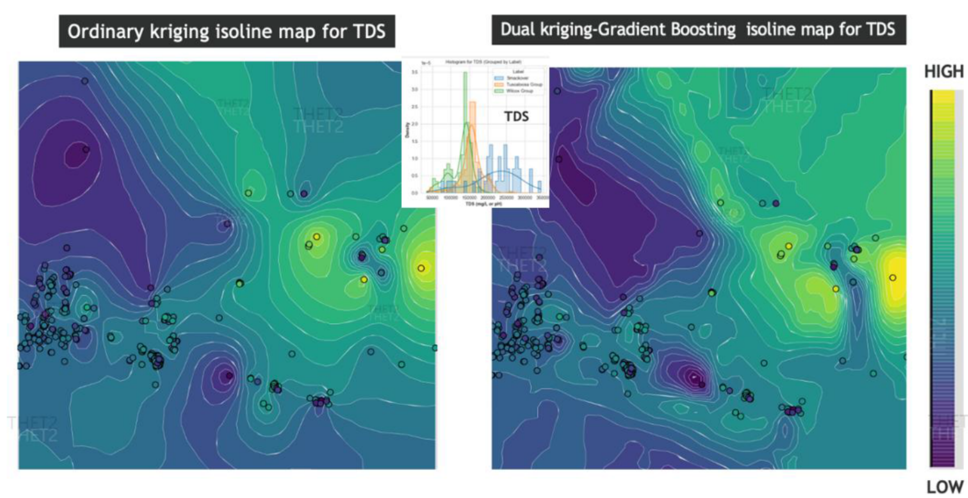

Figure 3—Comparison of prediction between ordinary kriging and dual kriging-gradient boosting prediction of TDS.

In Figure 3, the ordinary-kriging isoline map is driven solely by spatial autocorrelation of raw TDS measurements. As the stationary Gaussian process is a key kriging assumption, the interpolation smooths the surface uniformly. This phenomenon is manifested by the presence of large *bullseye* at high and low TDS areas with circular and symmetrical shapes, and subtle gradients are blurred around individual wells (monochromatic around well location). In practice, this points to the fact that isoline TDS maps could potentially mis-interpret or even miss narrow areas of high salinity that control corrosion, scaling, or other well management decisions. Ordinary kriging also does not have functionality to include non-spatial predictors such as formation identity or depth, that strongly influence TDS values across different stratigraphy (Smackover, Tuscaloosa and Wilcox groups). In contrast, the dual method map (right side of Fig. 3) ensures that gradient tree boosting (*XGBoostRegressor*) learns the nonlinear and formation dependent trends. They are captured visually in the middle-panel histogram, which shows very different TDS distributions for the Smackover, Tuscaloosa, and Wilcox groups. Universal kriging with external drift variables, is exclusively applied to the augmented input data, thus preserving local spatial continuity without forcing the data into a Gaussian distribution. The resulting contours of the dual isoline TDS map are sharper, elongated along structural trends, and better tied to clusters of wells in comparison to the ordinary-kriging isoline map. This corroborates the evidence that the hybrid method is honoring both geological controls and spatial structures. It is also visible that narrow, high TDS streaks appear where the ordinary map shows only gentle rises, while low-salinity pockets between wells are no longer over-smoothed, rather tortuous. For reservoir and groundwater management teams, this crisper isoline map translates into more reliable estimates of salinity at unsampled locations, which would indirectly influence better placement of monitoring wells, and impose tighter constraints on scaling inhibition or water treatment practices. In summary, coupled machine-learning method with kriging drift modelling could potentially bridges the gap between data driven pattern recognition and geostatistical interpolation, especially for geochemically realistic predictions wherever direct measurements are sparse.

Conclusion

A novel algorithm for a dual kriging-gradient tree boosting has been proposed to predict geochemical parameters, based on historical geochemical data of produced water samples at unsampled locations distant from well calibration points. A comprehensive analysis presented throughout this paper, supported by geostatistical and isoline maps, and endmember diagrams, demonstrating that the proposed methodology outperforms ordinary kriging in predicting complex geochemical parameters, such as TDS concentrations in formation water. Exploratory data analysis of the USGS geochemical dataset of produced water samples revealed a significant formation-dependent variability, non-Gaussian distribution, and a dominant Ca-Cl water type driven by evaporite dissolution, particularly in the Wilcox and Smackover groups. These patterns are not adequately captured by ordinary kriging due to inherent assumption of Gaussian stationarity, which leads to spatial over-smoothing and loss of key hydrogeochemical features, as shown in the isoline maps (Fig. 3). The dual method, on the other hand, incorporates nonlinear geochemical trends through machine learning (XGBoostRegressor) and enhances spatial accuracy through the usage of universal kriging with drift variables, resulting in more robust, and geologically realistic predictions, particularly at unsampled and new prospect locations. The observed improvement has highlighted the value of accurate geochemical fingerprinting not only for modeling salinity and scaling risk, but also for enhanced reservoir management, and produced water treatment across diverse geological settings. Similarly, an optimum combination of external drift variables could enhance the overall variance of the dual method. Despite several limitations, the proposed dual method is straightforward to implement by utilizing open-source python-based software components. The next step is to apply the dual kriging-gradient tree boosting method for operational field trials with a representative geochemical data base of produced water samples.

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